01.

Propane

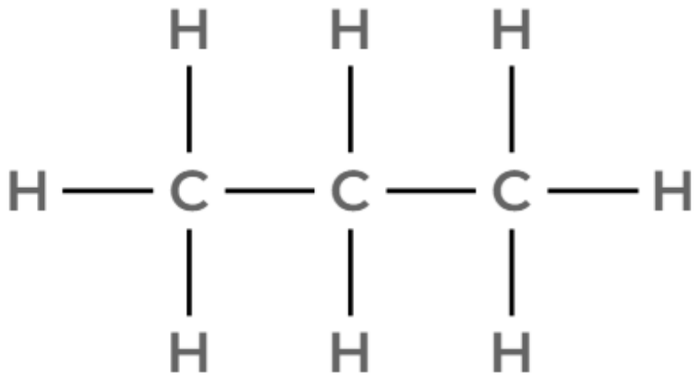
Common Name: Propane

Iupac Name: Propane

Molecular Formula

C_3H_8

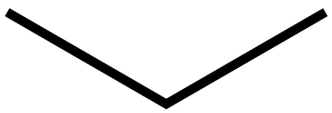
Lewis Structure



Condensed Structural Formula

$CH_3CH_2CH_3$

Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

02.

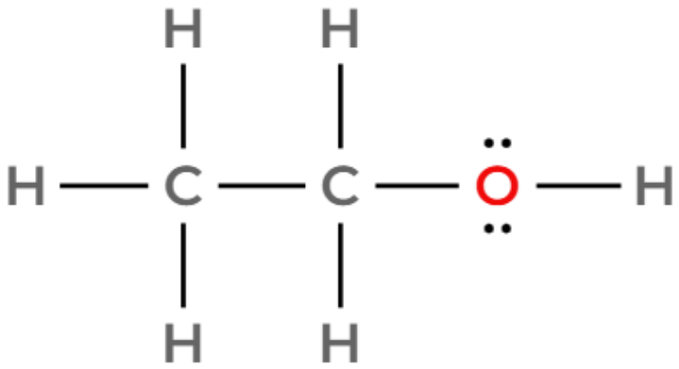
Ethanol

Common Name: Ethyl Alcohol
Iupac Name: Ethanol

Molecular Formula

C_2H_6O

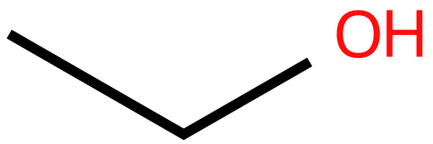
Lewis Structure



Condensed Structural Formula

CH_3CH_2OH

Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

03.

Acetone

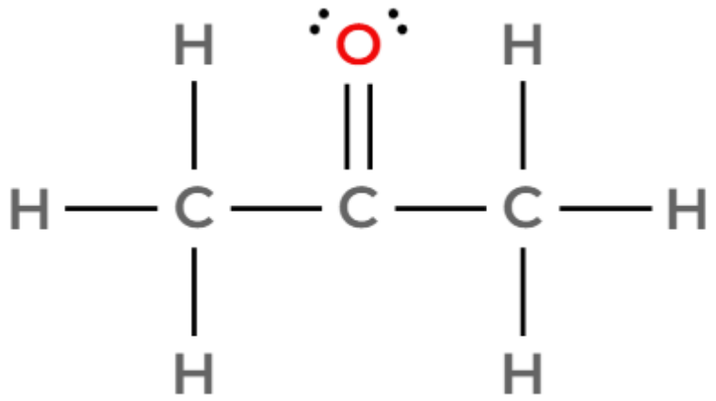
Common Name: Acetone

Iupac Name: Propan-2-one

Molecular Formula

C_3H_6O

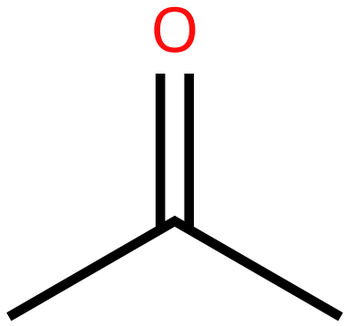
Lewis Structure



Condensed Structural Formula

CH_3CH_3OH

Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

04.

Acetic Acid

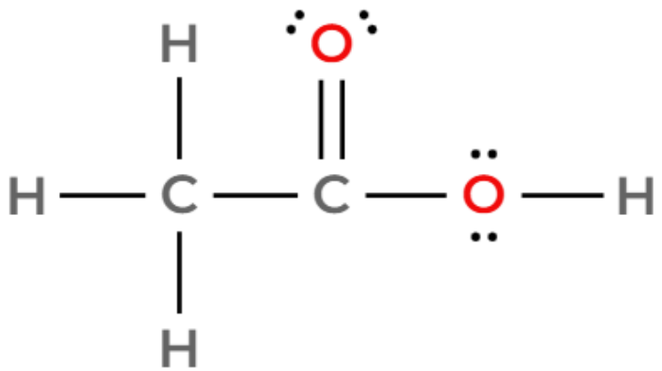
Common Name: Acetic Acid

Iupac Name: Ethanoic Acid

Molecular Formula

$C_2H_4O_2$

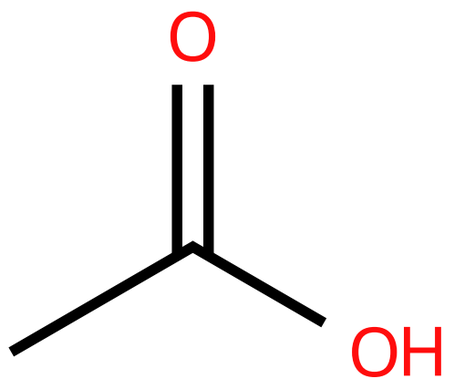
Lewis Structure



Condensed Structural Formula

CH_3COOH

Bond-Line Formula



- [x] atom count verified
- [x] valence checked
- [x] bond order checked
- [x] same connectivity confirmed

05.

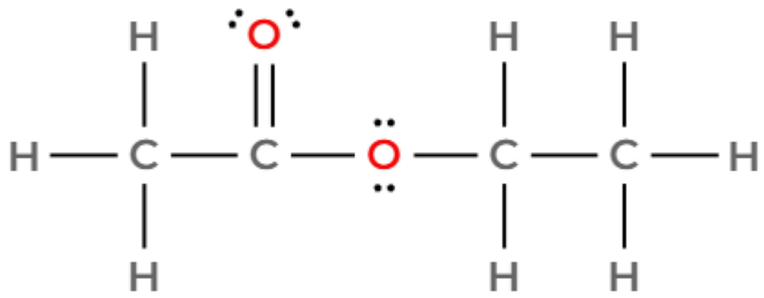
Ethyl Acetate

Common Name: Ethyl Acetate
Iupac Name: Ethyl Ethanoate

Molecular Formula

$C_4H_8O_2$

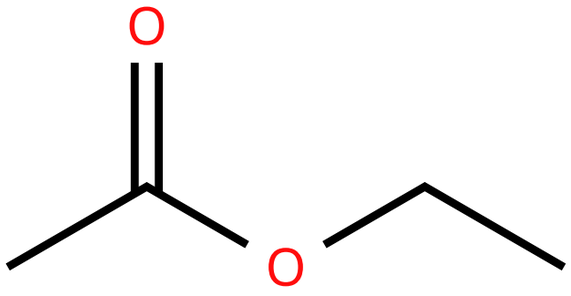
Lewis Structure



Condensed Structural Formula

$CH_3COOCH_2CH_3$

Bond-Line Formula



- [x] atom count verified
- [x] valence checked
- [x] bond order checked
- [x] same connectivity confirmed

06.

Cyclohexane

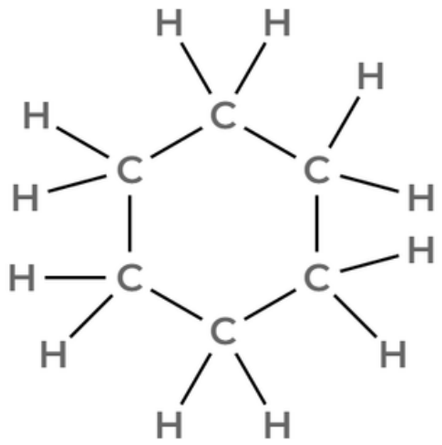
Common Name: Cyclohexane

Iupac Name: Cyclohexane

Molecular Formula

C_6H_{12}

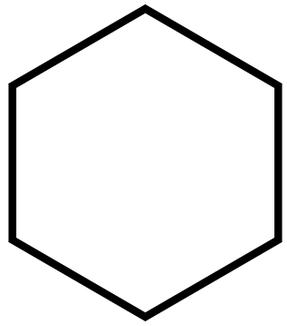
Lewis Structure



Condensed Structural Formula

$CH_2CH_2CH_2CH_2CH_2CH_2$

Bond-Line Formula



- [x] atom count verified
- [x] valence checked
- [x] bond order checked
- [x] same connectivity confirmed

07.

Cyclohexanone

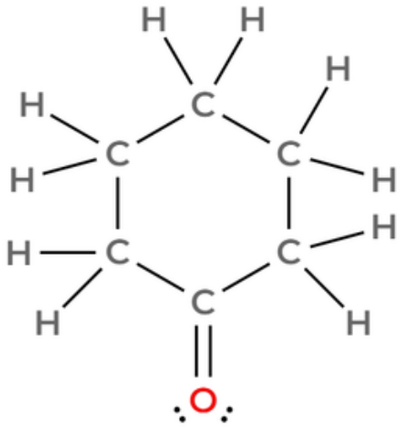
Common Name: Cyclohexanone

Iupac Name: Cyclohexanone

Molecular Formula

$C_6H_{10}O$

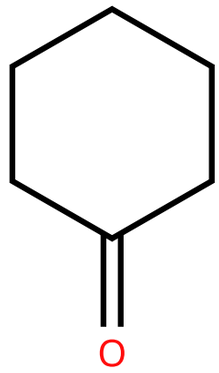
Lewis Structure



Condensed Structural Formula

$CH_2CH_2CH_2CH_2CH_2O$

Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

08.

Acetaminophen

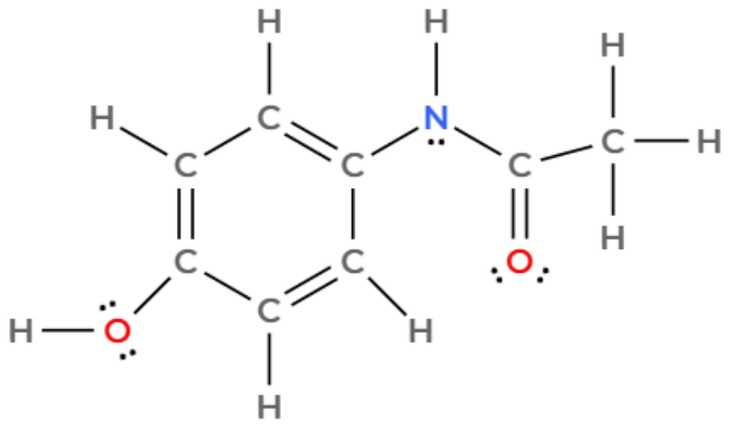
Common Name: Paracetamol

Iupac Name: N- (4 hydroxyphenyl)acetamide

Molecular Formula

$C_8H_9NO_2$

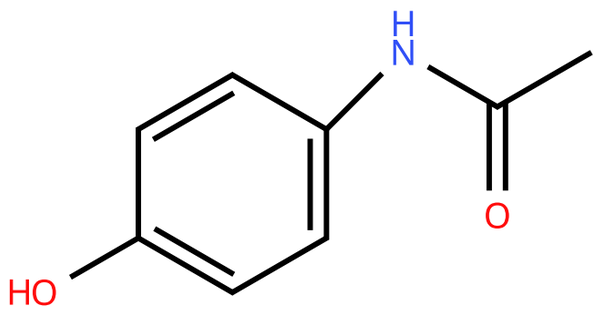
Lewis Structure



Condensed Structural Formula

$HOC_6H_4NHCOCH_3$

Bond-Line Formula



- [x] atom count verified
- [x] valence checked
- [x] bond order checked
- [x] same connectivity confirmed

09.

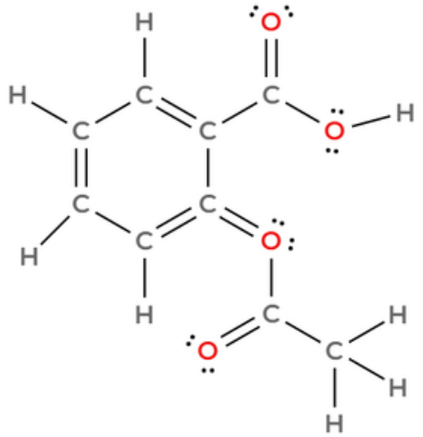
Aspirin

Common Name: Aspirin/ Acetylsalicylic acid
Iupac Name: 2-Acetoxybenzoic Acid

Molecular Formula

$C_9H_8O_4$

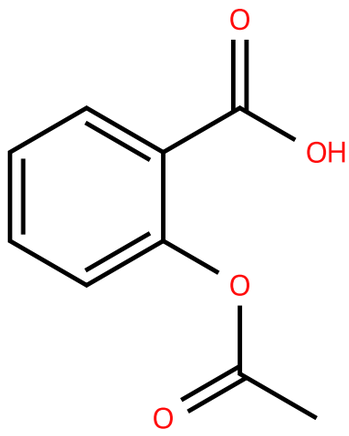
Lewis Structure



Condensed Structural Formula

$CH_3COOC_6H_4COOH$

Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed

10.

Caffeine

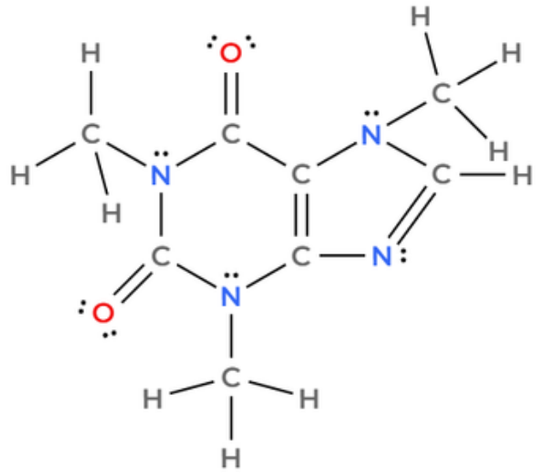
Common Name: Caffeine

Iupac Name: 1,3,7-Trimethyl-3,7-dihydro-1H-purine-2,6-dione

Molecular Formula

$C_8H_{10}N_4O_2$

Lewis Structure

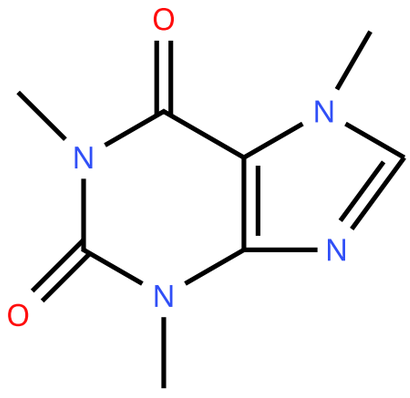


Condensed Structural Formula

$C_8H_{10}N_4O_2$

Note: Caffeine has no condensed structural formula because it consists of a complex fused bicyclic ring system.

Bond-Line Formula



☑ atom count verified ☑ valence checked ☑ bond order checked ☑ same connectivity confirmed